✏■ Resolution Exercises (Propositional Logic)

Exercise 1: Multiple Implications and Negations

Knowledge Base (KB):

1. $(P \lor Q) \rightarrow R$
2. $R \rightarrow (S \land T)$
3. $\neg S \lor U$
4. Q

Conclusion: U

Tasks:

1. Eliminate implications.
2. Push negations inward (e.g., $\neg(A \land B) \equiv \neg A \lor \neg B$).
3. Convert into CNF using distributive laws.
4. List all clauses.
5. Add the negation of the conclusion.
6. Apply resolution until you derive a contradiction.

Exercise 2: Biconditional + Nested Structures

Knowledge Base (KB):

1. $A \leftrightarrow (B \lor C)$
2. $A \rightarrow D$
3. $C \rightarrow E$
4. $\neg E \lor F$

Conclusion: D

Tasks:

1. Break the biconditional into two implications.
2. Eliminate all implications.
3. Convert into CNF (apply distributivity carefully).

4. List all clauses.

5. Add the negation of the conclusion.

6. Use resolution to try to reach a contradiction.

Exercise 3: Mixing Implications, Negations, and Distributivity

Knowledge Base (KB):

1. $(M \land N) \rightarrow P$

2. $P \rightarrow (Q \lor R)$

3. $\neg Q \lor S$

4. $\neg R \lor T$

5. M

6. N

Conclusion: S

Tasks:

1. Rewrite implications.

2. Push negations inward.

3. Put formulas into CNF (watch out for distributivity when expanding).

4. Collect the clauses.

5. Add the negation of the conclusion.

6. Apply resolution systematically to derive a contradiction.